Climate Change, Stock Productivity, and Fishing Effort in Chile's Multi-Species Small Pelagic Fishery

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Abstract

We study how climate change will affect stock productivity and fishing effort in Chile's multi-species small pelagic fishery, composed of anchoveta (*Engraulis ringens*), jack mackerel (*Trachurus murphyi*), and sardine (*Strangomera bentincki*). We estimate a three-equation Seemingly Unrelated Regression (SUR) model for stock growth as a function of biomass, sea surface temperature, and chlorophyll-a, and a negative binomial model for annual fishing trips as a function of prices, quotas, weather, and regulatory closures. Using CMIP6 projections from the IPSL-CM6A-LR model under SSP2-4.5 and SSP5-8.5 scenarios, we apply the delta method to project changes in environmental conditions at mid-century (2041–2060) and end-of-century (2081–2100). We decompose the projected climate impact on fishing effort into a direct weather channel (wind affecting operational days) and an indirect biomass channel (temperature and chlorophyll affecting stock productivity and harvest allocations). Results reveal a stark asymmetry: the artisanal fleet benefits from warming-driven sardine expansion (effort increases of 20–250%), while the industrial fleet faces declining jack mackerel productivity (effort declines of approximately 22%). The indirect biomass channel dominates, with the direct weather effect contributing less than 5% of the total response. These findings highlight heterogeneous climate vulnerability across fleet segments and have implications for quota allocation policy in multi-species fisheries.

1 Introduction

The distribution and abundance of marine resources are changing in response to environmental conditions such as global ocean warming (Poloczanska et al., 2013). Climate change is projected

to shift species distributions, reducing availability in some areas while increasing it in others, with direct consequences for fishing communities (Sumaila et al., 2011). Understanding how these changes will propagate to fishing activity is a prerequisite for designing effective adaptation policies. Adaptation responses by fishers are heterogeneous and depend on institutional, economic, and ecological context (Jardine et al., 2020; Zhang & Smith, 2011), which motivates empirical analyses at the local level.

The Chilean small pelagic fishery (SPF) provides a particularly informative setting. It is the largest fishery in the country, accounting for almost 94% of total Chilean landings in 2019 (SUBPESCA, 2020), and is dominated by three species—anchoveta (*Engraulis ringens*), sardine (*Strangomera bentincki*), and jack mackerel (*Trachurus murphyi*)—that share habitat and are coupled through both trophic and market linkages. Cheung et al. (2010) project Chile among the top fishing nations in terms of declines in maximum catch potential by mid-century, with an estimated reduction of 6–13% under the SRES A1B scenario. Yet the Southeast Pacific remains one of the least studied regions regarding the economic impacts of climate change on fisheries (Sumaila et al., 2011), and, to our knowledge, no existing study for Chile combines multi-species stock dynamics with vessel-level effort responses under projected climate scenarios.[1]

A multi-species framework is essential in this setting because the three target species are ecologically and economically interdependent. Jack mackerel preys on anchoveta and sardine, so reductions in anchoveta or sardine availability may reflect not only direct environmental drivers but also changes in predation pressure (Alheit & Niquen, 2004; Arancibia et al., 2019). Together with a fleet structure in which the industrial and artisanal segments target different species, this generates cross-fleet spillovers through biology, quota allocation rules, and shared port infrastructure. Current management—which sets the TAC on a single-species basis—does not internalize these interdependencies, strengthening the case for a multi-species approach.

We contribute to the literature in three ways. First, we jointly estimate multi-species stock dynamics and fleet-level fishing effort for Chile's Centro-Sur SPF, combining a three-equation Seemingly Unrelated Regression (SUR) system for interannual biomass growth—augmented with sea surface temperature (SST) and chlorophyll-a—with a negative binomial model for annual fishing trips estimated separately for the industrial and artisanal fleets. Second, we link the estimated econometric models to CMIP6 climate projections (IPSL-CM6A-LR) under SSP2-4.5 and SSP5-8.5 at mid-century (2041–2060) and end-of-century (2081–2100), using the delta method to translate projected environmental changes into effort responses. Third, we decompose the projected climate impact on fishing effort into a *direct weather channel*—changes in wind affecting operational feasibility—and an *indirect biomass channel*—changes in temperature and chlorophyll affecting stock productivity and vessel-level harvest allocations. The decomposition isolates the dominant pathway through which climate change reshapes fishing activity in this fishery and highlights het-

[1] For Chile, the only empirical study of fisher behavior using discrete choice modeling that we are aware of is @Pena-Torres2017-gn, which analyses how the El Niño–Southern Oscillation (ENSO) affects location choices in the jack mackerel fishery.

erogeneous vulnerability across fleet segments, with direct implications for how quota allocation policy should adapt.

2 The small pelagic fishery in Chile

The Chilean small pelagic fishery is structured into two latitudinal zones with distinct species composition and fleet activity. In the northern zone competition is largely between anchoveta and jack mackerel, while in the Central-South region (Regions V–X and XIV) all three species play a major role. The Central-South is therefore the most relevant setting for the study of species interactions and potential substitution within a multispecies management framework, and it is the focus of this paper.

The jack mackerel fishery was initially concentrated in northern Chile but shifted to Central-South Chile in the mid-1980s, where the main fishing grounds have traditionally operated within 50 nautical miles of the coast (Peña-Torres et al., 2017). Landings of the three species have historically been destined primarily for the reduction industry: between 1987 and 2004, approximately 85% of jack mackerel landings were processed into fishmeal and fish oil (Peña-Torres et al., 2017). Today, the main ports servicing the SPF include San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral. Sardine and anchoveta are harvested as a mixed fishery in the Central-South region: the two species share the same habitat and are caught with the same gear, making differentiation at the point of capture effectively impossible (Dresdner et al., 2013).

2.1 Stock status

Central-South anchoveta was classified as collapsed through 2018, shifted to overexploited status in 2019, and has been fished within maximum sustainable yield (MSY) limits since 2020. Sardine has generally remained within MSY limits, except in 2021 and 2023 when it was classified as overexploited. Jack mackerel was overexploited until 2018 and has since been harvested within MSY limits.

2.2 Management regime

The Chilean SPF is managed through an annual Total Allowable Catch (TAC, *Cuota Global*) divided between the industrial and artisanal sectors, with small shares reserved for research and contingency. The TAC is subdivided by region and season, and unused quotas can be reassigned within the fishing year. Anchoveta and sardine are regulated as a mixed-species fishery—each species has its own quota, but substitution between them is permitted.

Since 2013, the industrial sector has operated under an individual transferable quota (ITQ) system known as Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP). Class A

licenses were allocated on the basis of historical catches, while Class B licenses (up to 15% of the industrial fraction) are allocated through sealed-bid first-price auctions that began in 2015. Peña-Torres et al. (2022) document that these auctions have exhibited low participation, limited ability to reflect economies of scale, and signs of coordinated bidding. The artisanal sector operates under a regulated freedom-to-fish regime, with regional TACs allocated through the *Régimen Artesanal de Extracción* (RAE) by area, vessel size, landing site, or fisher organization.

Biological closures (*vedas*) are a central regulatory instrument for sardine and anchoveta. In the Central-South region, reproductive closures run from July–October in regions IX–XIV and from August–October in regions V–VIII, while recruitment closures run from January to February across all regions. In total these yield 151 closed days per year in regions V–VIII and 182 closed days in regions IX–XIV. Jack mackerel has no seasonal biological closures.

3 Data and methodology

Our empirical approach has three stages. First, we estimate a multi-species stock dynamics model using a SUR framework to capture how environmental conditions (SST, chlorophyll-a) affect the growth capacity of each species. Second, we estimate a negative binomial model for annual fishing trips as a function of prices, harvest allocations, weather conditions, and regulatory closures, separately for the artisanal and industrial fleets. Third, we combine the estimated models with CMIP6 climate projections to assess how climate change will affect stock productivity and fishing effort through both direct (weather) and indirect (biomass) channels. A companion paper extends this framework to include trip-level cost functions, an inverse demand system, and full numerical optimization of quota paths following Kasperski (2015).

3.1 Historical data

We use data requested from the Chilean Fisheries Development Institute (Instituto de Fomento Pesquero, IFOP) covering the 2013–2024 period. The dataset includes trip-level microdata, which contain detailed records on vessel identifiers, departure and arrival times, vessel capacity, fleet and gear type, ports of departure and landing, fishery codes, haul timing and location, species composition, retained catch, and trip activity. In addition, we requested annual information on stock abundance and vessel landings by port, county, region, country, and species. Finally, we use ex-vessel prices, reported monthly or annually by port, county, region, country, and species. These prices reflect those paid by processing plants to fishers at the point of first sale and are obtained through IFOP's landing surveys, which do not necessarily cover all market transactions.

For the environmental covariates, we use data from the E.U. Copernicus Marine Service Information, accessed through the Copernicus Marine Toolbox API. Salinity, sea surface temperature,

and current speed and direction were obtained from the Global Ocean Physics Reanalysis (GLORYS12V1), which provides data at a 1/12° horizontal resolution with 50 vertical levels (E.U. Copernicus Marine Service Information, 2025c). Wind speed and direction at the surface were obtained from the Global Ocean Hourly Reprocessed Sea Surface Wind and Stress from Scatterometer and Model dataset, available at 0.125° horizontal spatial resolution and hourly frequency (E.U. Copernicus Marine Service Information, 2025b). Chlorophyll-a concentrations were obtained from the Global Ocean Colour dataset, which provides data at ~4 km horizontal resolution (E.U. Copernicus Marine Service Information, 2025a). All environmental data were retrieved daily (hourly in the case of winds) for the 2013–2024 period, covering the Chilean Exclusive Economic Zone (EEZ) between 32°S and 41°S (Figure 1)

Imputation of missing jack mackerel biomass To address the years in which no acoustic survey was conducted in south-central Chile—or in which insufficient information was available to produce a stock assessment—we imputed missing jack mackerel biomass values. Of the 25 years in the sample period (2000–2024), jack mackerel CS biomass is directly observed in 17 years; the remaining 8 years require imputation. The primary predictor is jack mackerel biomass from the Northern zone, estimated from independent acoustic surveys conducted since 2010 (correlation with observed CS biomass: $r = 0.82$, $N = 7$ overlapping years).

We estimated five alternative GLM specifications with a Gamma distribution and log link, ranging from a parsimonious model with Northern biomass as the sole predictor (2 parameters) to a saturated specification including Northern biomass, its square, the SPRFMO spawning biomass index, and their interaction (5 parameters). All models using Northern biomass are estimated with only 7 observations (years where both CS and Norte are simultaneously observed). The most complex specification exhibited severe overfitting (pseudo-$R^2 = 0.97$, but biologically implausible predictions requiring ad hoc censoring in two years). We therefore selected a parsimonious specification—Model B: Northern biomass and its square (3 parameters)—which achieves an in-sample pseudo-$R^2 = 0.90$ and a leave-one-out cross-validation $R^2 = 0.78$, indicating acceptable out-of-sample predictive performance. For the single year where Northern biomass is also unavailable (2022), we use linear interpolation between adjacent observed values as a fallback.

The SPRFMO spawning biomass index, which covers the transzonal stock across the EEZ of Chile, Ecuador, and Peru, was also considered as a predictor. However, its correlation with observed CS biomass is low ($r = 0.11$, $N = 17$), reflecting the weak correspondence between regional stock dynamics and local abundance in south-central Chile. The interpolated series, together with a source flag indicating whether each observation is acoustically observed, GLM-predicted, or linearly interpolated, is used in subsequent estimation. To assess sensitivity to the imputation procedure, we report robustness checks using only observed jack mackerel biomass (Appendix A).

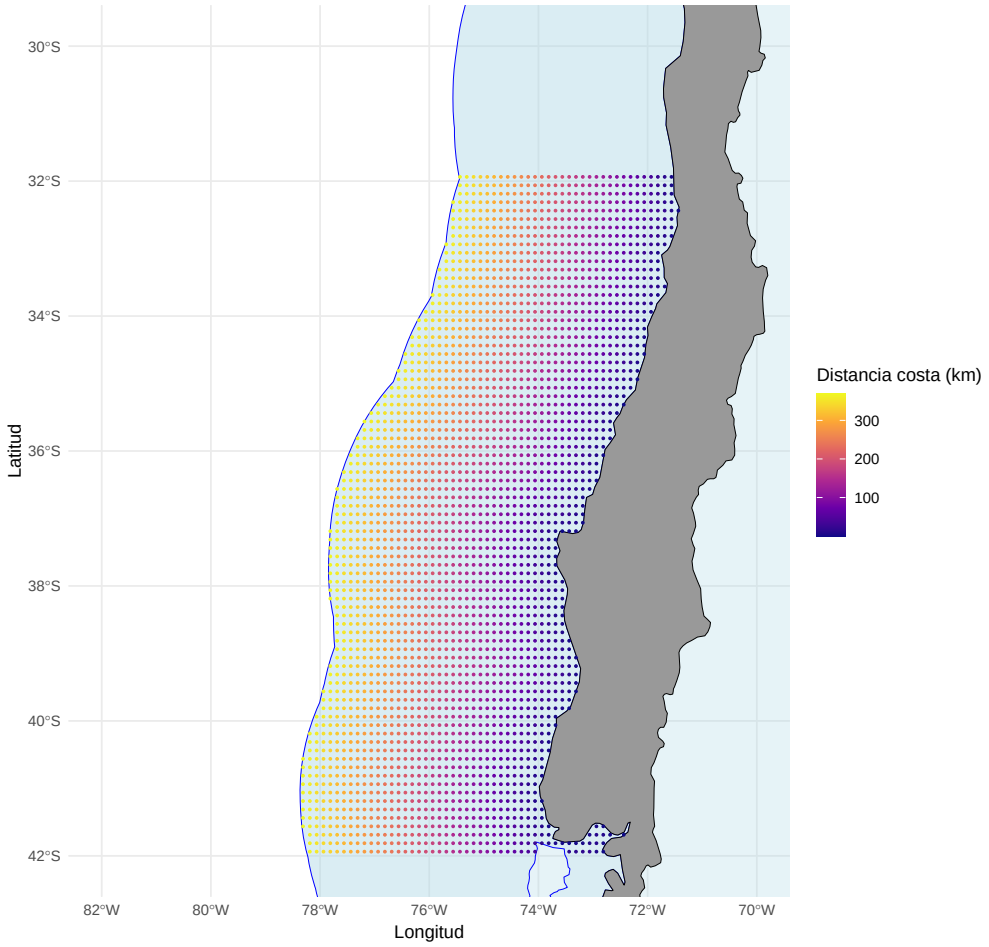

Figure 1: Geographical extent of the study area used for environmental covariates limited to the Chilean Exclusive Economic Zone

3.2 Future climate data

Climate projections for SST, chlorophyll-a, and near-surface wind speed are obtained from the CMIP6 archive, using the IPSL-CM6A-LR Earth system model. We use monthly outputs for two future scenarios—SSP2-4.5 (moderate) and SSP5-8.5 (high emissions). The projection horizons, baseline period, and delta-method procedure used to translate CMIP6 outputs into our empirical models are presented in Section 3.4.

3.3 Econometric models

3.3.1 Stock dynamics

To estimate stock dynamics, we use annual data on species-specific biomass (stock abundance) and vessel landings within Chile's Exclusive Economic Zone (EEZ). Following Kasperski (2015),

the baseline model for the interannual growth of each species i is represented by a discrete logistic function that accounts for intra- and inter-species interactions:

$$x_{i,y+1} + h_{iy} = \underbrace{(1+r_i)x_{iy} + \eta_i x_{iy}^2}_{R_i(x_{iy})} + \underbrace{\sum_{j \neq i}^{n-1} a_{ij} x_{iy} x_{jy}}_{I_i(x_y)} \quad i = 1, \dots, n \tag{1}$$

where x_{iy} is the biomass of species i in year y, n is the total number of species, h_{iy} is the annual harvest, r_i is the intrinsic growth rate, η_i is the density-dependent parameter, and α_{ij} captures pairwise species interactions. The system of n growth equations is estimated simultaneously using Seemingly Unrelated Regression (SUR). Following Richter et al. (2018), we augment (1) by including environmental covariates Env_{iy} that affect stock dynamics, along with an error term ε_{iy} representing stochastic recruitment:

$$x_{i,y+1} + h_{iy} = \underbrace{(1+r_i)x_{iy} + \eta_i x_{iy}^2}_{R_i(x_{iy})} + \underbrace{\sum_{j \neq i}^{n-1} a_{ij} x_{iy} x_{jy}}_{I_i(x_y)} + \rho_i Env_{iy} + \varepsilon_{iy} \quad i = 1, \dots, n \tag{2}$$

where ρ_i denotes the environmental response parameters. Each species has its own equation, allowing for species-specific growth, density dependence, and environmental sensitivities.

Environmental conditions were summarized annually using sea surface temperature (SST) and chlorophyll-a concentration (CHL), both averaged over the South-Central Chile region within the EEZ. These variables were selected due to their recognized influence on small pelagic productivity and spatial distribution (Axbard, 2016). SST serves as a proxy for large-scale oceanographic regimes (warm vs. cold phases) that shape recruitment success along the Humboldt Current System. Anchoveta typically dominates during cold, nutrient-rich phases, whereas sardine tends to increase under warmer regimes (Cahuin et al., 2009; Yáñez et al., 2014). CHL reflects interannual variation in primary productivity, which forms the energetic base sustaining small pelagics (see Cheung et al., 2008; Jennings et al., 2008).

To select the preferred specification, we compared five candidate models by AIC and BIC, varying the inclusion of quadratic environmental terms, cross-species interaction terms, and wind covariates (Table 1). Wind effects were excluded first, as a joint exclusion test failed to reject their omission. Cross-species interaction terms ($x_{iy}x_{jy}$) and quadratic CHL were subsequently tested but did not improve fit—both AIC and BIC increased when these terms were added. The preferred specification retains own biomass, its square, SST, SST^2, and CHL (linear) for each species, yielding six parameters per equation and approximately 17 residual degrees of freedom with $N = 23$ observations. This parsimonious model is estimated as:

$$x_{i,y+1} + h_{iy} = (1 + r_i)x_{iy} + \eta_i x_{iy}^2 + \rho_{i,1} SST_y + \rho_{i,2} SST_y^2 + \rho_{i,3} CHL_y + \varepsilon_{iy} \quad i = 1, \dots, n \tag{3}$$

where all environmental variables are centered at their sample means to reduce collinearity between linear and quadratic terms.

Figure 2 shows that anchoveta, sardine, and jack mackerel exhibit distinct biomass trajectories over time, with no evidence of strong interannual co-movement across species. Biomass levels differ markedly between them, and the figure suggests only limited interrelation alongside clear divergences in year-to-year dynamics. Harvest pressure is also visibly associated with these fluctuations; for instance, jack mackerel experienced an abrupt decline during the late 2000s, consistent with the combined effects of intense fishing pressure and unfavorable environmental conditions.

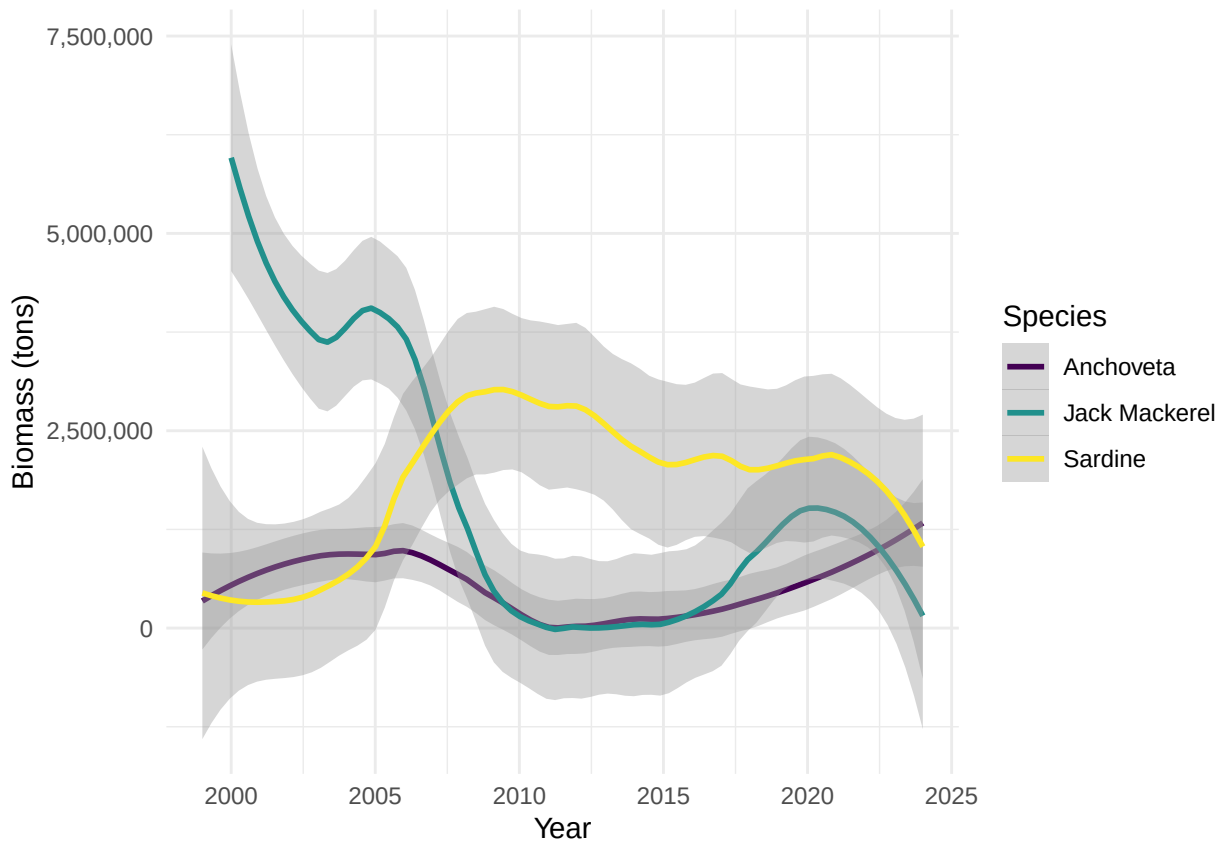

Figure 2: Estimated biomass of small pelagic species in Chile (2000–2024)

The SUR framework is particularly suited to this multi-species setting, as it allows for contemporaneous correlation among residuals that naturally arise from shared non-observed environmental shocks, trophic linkages, and imperfectly observed ecological processes. We estimate a three-equation SUR system using the SEM framework implemented in lavaan, freely allowing the error terms across biomass equations to be correlated. The model is estimated via robust maximum like-

lihood (MLR), yielding SUR coefficients with Huber–White robust standard errors. Cross-species interaction terms were tested as an extension but did not improve model fit (see Section 4.1 and Appendix A).

3.3.2 Total annual trips

We model the annual number of fishing trips taken by vessel v in year y as a count process following Kasperski (2015). Since the available logbook data only record purse-seine operations, the unit of observation is a vessel–year. Separate effort models are estimated for the industrial and artisanal fleets, allowing fishing activity to respond differently to economic conditions and regulatory constraints across sectors. This specification explicitly accommodates technological heterogeneity across fleet segments, reflecting differences in vessel capacity, operating scale, and production technology.

The baseline specification is a Poisson model:

$$T_{vy} \sim \text{Poisson}(\lambda_{vy}), \qquad \lambda_{vy} = \exp\big(U'_{vy}\beta\big)\,, \tag{4}$$

where T_{vy} denotes the total number of purse-seine trips recorded for vessel v in year y. Given substantial overdispersion in the data—the variance-to-mean ratio of T_{vy} is 22.4 for the artisanal fleet and 4.9 for the industrial fleet—we estimate a negative binomial (NB) model, which nests the Poisson as a special case. A likelihood ratio test strongly rejects the Poisson restriction in both fleets ($p < 0.001$).

The vector of explanatory variables U_{vy} includes output prices by species, vessel-level allocated harvest, fixed vessel characteristics, and operating conditions:

$$U_{vy} = [p_{sy},\, H_{vy}^{alloc},\, Z_v,\, Env_{vy}]. \tag{5}$$

Output prices p_{sy} correspond to species-specific ex-vessel prices paid by processing plants to fishers, obtained from IFOP's manufacturing survey and deflated to constant 2018 pesos using the consumer price index from the Central Bank of Chile. Consistent with Dresdner et al. (2013), prices are measured as annual averages over peak fishing months to reflect the economically relevant conditions faced by vessels when planning annual effort.

Because the objective of the model is to characterize vessels' effort responses to changes in prices, quotas, and environmental conditions within the small pelagic fishery, we restrict attention to SPF trips recorded in logbooks. These trips account for approximately 95% of observed fishing revenue and therefore capture the primary economic margin through which vessels adjust annual fishing effort.

Quota shares do not enter Eq.~(4) directly. Instead, shares translate annual TACs into vessel-level allocated harvests. Let ω_{vs}^{r} denote vessel v's historical share of landings of species s within its administrative region r (for artisanal vessels) or regulatory zone (for industrial vessels), computed over the full sample period. Given the effective TAC $\bar{Q}_{sy}^{r}$ for species s in year y in region r—obtained from SERNAPESCA quota monitoring records and reflecting all inter-fleet transfers and adjustments—vessel-level allocated harvest is:

$$H_{vy,s}^{alloc} = \omega_{vs}^{r}\,\bar{Q}_{sy}^{r}, \qquad H_{vy}^{alloc} = \sum_{s} H_{vy,s}^{alloc}. \tag{6}$$

Each vessel is assigned to its administrative region based on its modal departure port from the logbook records. Artisanal TACs are assigned at the regional level (regions V, VIII, IX, XIV, X), while industrial TACs follow the regulatory zone structure established by SUBPESCA (zones V–IX and XIV–X for jack mackerel; zone V–X for sardine and anchoveta). This regional construction introduces cross-sectional variation in H_{vy}^{alloc} that reflects the heterogeneous quota environments faced by vessels operating in different parts of the Centro-Sur fishery.

The vector Z_v contains time-invariant vessel characteristics, including hold capacity (log of cubic meters) and vessel type. Hold capacity determines the physical constraint on catch per trip and proxies for vessel scale.

The vector Env_{vy} captures operating and regulatory constraints that vary across both vessels and years. Unlike the environmental variables in the stock dynamics model, which reflect biological productivity (SST, chlorophyll-a), the variables in the trip equation capture conditions that affect the *feasibility* of fishing operations. Two vessel–year variables are constructed using each vessel's center of gravity (COG)—the catch-weighted centroid of its haul locations over the sample period. First, the number of days with adverse weather conditions is computed as the count of days per year in which wind speed exceeds 8~m/s at the environmental grid point nearest to the vessel's COG.[2] Second, the number of days closed due to biological closures (*vedas*) is assigned based on the regulatory zone corresponding to the vessel's COG latitude. In the Centro-Sur region, sardine and anchoveta face seasonal closures for reproduction (August–October in regions V–VIII; July–October in regions IX–XIV) and recruitment (January–February in all regions), yielding 151 and 182 closed days per year, respectively. Jack mackerel has no biological closures.

Quota prices are not included explicitly in the trip equation. Instead, the economic scarcity of quota is captured implicitly through aggregate TAC levels and vessel-level allocated harvest.

Trips, harvest, and prices may be jointly determined within the year, implying potential endogeneity in reduced-form trip regressions. However, the purpose of Eq.~(4) is not causal identification but rather to provide an empirically grounded behavioral relationship that maps changes in prices,

[2]The wind threshold of 8 m/s corresponds approximately to Beaufort scale 5, at which purse-seine operations become difficult for artisanal vessels. This threshold was selected based on model fit (AIC comparison across 8, 10, and 12 m/s thresholds); robustness checks are reported in Appendix A.

quotas, and environmental conditions into expected fishing effort under projected climate scenarios.

3.4 Projection approach

We assess the impact of climate change on fishing effort through two channels: (i) a *direct weather channel*, where projected changes in wind speed alter the number of bad weather days that constrain fishing operations, and (ii) an *indirect biomass channel*, where projected changes in sea surface temperature and chlorophyll-a affect stock growth capacity through the SUR equations, which in turn modifies harvest allocations and fishing effort through the trip equation.

Climate projections are derived from the IPSL-CM6A-LR Earth system model under two Shared Socioeconomic Pathways: SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions). We apply the delta method (Burke et al., 2015; Free et al., 2019), which preserves the observed interannual variability and fine-scale spatial structure from satellite records while imposing the climate change signal from CMIP6. For each variable, the delta is computed as the difference (additive for SST and wind speed) or ratio (multiplicative for chlorophyll-a) between the CMIP6 future climatology and the CMIP6 historical climatology (1995–2014 baseline), applied cyclically to each calendar month. We consider two future time windows: mid-century (2041–2060) and end-of-century (2081–2100).

For the direct weather channel, monthly wind speed deltas from CMIP6 are interpolated to the observed 0.125-degree wind grid and applied additively to daily wind speed observations. Bad weather days (wind speed exceeding 8 m/s) are then recomputed under each scenario and fed into the NB trip equation. For the indirect biomass channel, we compute changes in growth capacity for each species using comparative statics on the SUR environmental coefficients, evaluating the predicted change in y_i (next-period biomass plus current harvest) under projected versus historical SST and chlorophyll-a, holding biomass at its historical mean. These growth capacity changes are translated into proportional changes in harvest allocations (H_{vy}^{alloc}), weighted by each vessel's historical species composition.

As noted by Kasperski (2015), the multi-species bioeconomic framework is particularly suited for long-term projections of fish populations and fishing activity rather than intra-annual management responses. However, because our empirical models are estimated on interannual variability rather than decadal trends, the projections presented here capture the impact of changed environmental conditions without accounting for endogenous adaptation by fishers or managers (Aufhammer, 2018). This provides a useful benchmark for understanding the direction and magnitude of climate impacts, while acknowledging that behavioral responses may attenuate or amplify these effects over time.

4 Results

4.1 Stock biomass model

Table 1 summarizes the model selection exercise. Five candidate specifications were compared, ranging from a baseline with only own-biomass terms to a full model including quadratic chlorophyll-a and cross-species interaction terms. The preferred specification—which includes own biomass, its square, SST, SST^2, and chlorophyll-a—yields the lowest AIC and BIC. Adding quadratic CHL terms or cross-species interactions does not improve fit; in fact, both AIC and BIC increase when these terms are included. Wind variables were excluded in an earlier specification search, as a joint exclusion test failed to reject their omission. The preferred specification contains six parameters per equation, yielding approximately 17 residual degrees of freedom with $N = 23$ observations.

Table 2 reports the SUR estimates of annual biomass dynamics for sardine, anchoveta, and jack mackerel. The system is estimated via robust maximum likelihood (MLR), yielding Huber–White standard errors and allowing unrestricted contemporaneous correlation across equations.

All three species exhibit strong interannual persistence and significant density dependence, consistent with discrete logistic growth. However, the species differ markedly in their sensitivity to environmental conditions.

Sardine biomass displays the strongest environmental response. Chlorophyll-a enters positively and is highly significant ($\hat{\rho}_{CHL} = 80.99$, $p < 0.001$), indicating that years with higher primary productivity favor sardine growth. The quadratic SST term is also significant and positive ($\hat{\rho}_{SST^2} = 54.74$, $p = 0.001$), suggesting a convex response to temperature anomalies—sardine biomass increases more sharply under larger SST deviations from the mean. This pattern is consistent with evidence from the Humboldt Current System, where sardine tends to exhibit smoother population changes associated with warmer or transitional oceanographic regimes (Cahuin et al., 2009; Yáñez et al., 2014).

Anchoveta biomass is dominated by density dependence, with the strongest own-biomass coefficient ($\hat{r} = 1.11$, $p < 0.001$) and a highly significant quadratic term ($\hat{\eta} = -0.202$, $p < 0.001$). SST has a marginally significant negative effect ($p = 0.087$), but neither SST^2 nor CHL are significant. This asymmetric environmental response—where anchoveta reacts primarily to temperature rather than productivity—is consistent with the ecological literature. Anchoveta in the Humboldt Current is known to respond sharply to short-term cold, nutrient-rich conditions, dominating during cold phases of decadal-scale oceanographic variability (Alheit & Niquen, 2004; Cahuin et al., 2009; Yáñez et al., 2014).

Jack mackerel shows strong persistence ($\hat{r} = 1.06$, $p < 0.001$) but a distinct environmental signature. The quadratic SST term is large and highly significant ($\hat{\rho}_{SST^2} = -56.49$, $p = 0.001$), indicating a clear thermal optimum: biomass declines when temperatures deviate substantially from

the long-run mean in either direction. Neither linear SST nor CHL are individually significant, suggesting that jack mackerel growth is less sensitive to annual fluctuations in upwelling productivity than sardine. This weaker coupling to coastal environmental variability is consistent with jack mackerel's broader spatial distribution, deeper foraging behavior, and transboundary stock structure within the Southeast Pacific (Peña-Torres et al., 2017).

These contrasting environmental responses have direct implications for climate projections. Under warming scenarios, sardine may benefit from increased thermal variability if accompanied by sustained productivity, whereas anchoveta could face reduced biomass if warming erodes the cold, nutrient-rich conditions it depends on. Jack mackerel, with its thermal optimum response, would be adversely affected by both warming and cooling extremes.

Cross-species interactions were tested but did not improve model fit (Table 1). The residual covariances across equations are also statistically insignificant, suggesting that shared unobserved shocks—beyond those captured by SST and CHL—do not generate strong contemporaneous comovement across species. This does not rule out ecological interactions (e.g., predator–prey dynamics between jack mackerel and small pelagics), but it indicates that such effects are not detectable in the annual aggregate biomass data given the available sample size.

Appendix A presents three robustness checks. Table 6 restricts the jack mackerel series to observed (non-interpolated) years, reducing the sample to 12 observations and requiring a more parsimonious specification without SST^2. Table 7 estimates a two-species SUR for sardine and anchoveta only, treating jack mackerel biomass as an exogenous covariate. Table 8 augments the main specification with cross-species interaction terms, confirming that their coefficients are individually and jointly insignificant. The core findings—strong density dependence, positive CHL effects on sardine, and a thermal optimum for jack mackerel—are qualitatively robust across all specifications, lending confidence that the results are not driven by the interpolation procedure.

Table 1: Model selection for SUR biomass dynamics

Specification	K/eq	AIC	BIC	ΔAIC
No environment	3	490.7	507.7	13.1
Main (SST + SST2 + CHL)	6	477.6	504.9	0.0
Main + CHL2	7	482.9	513.6	5.3
Main + interactions	8	485.0	519.1	7.4
Full (interactions + CHL2)	10	490.5	528.0	12.9

4.2 Total Annual trips

Table **??** reports the negative binomial estimates separately for each fleet segment.

% Error: Unrecognized object type. % Error: Unrecognized object type.

Table 2: SUR estimates of biomass dynamics (main specification, N = 23)

	Dependent variable:		
	Sardine	Anchoveta	Jack mackerel
	(1)	(2)	(3)
Own biomass (x_{it})	0.438^{***} (0.113)	1.112^{***} (0.157)	1.057^{***} (0.137)
Own biomass^2 (x_{it}^2)	-0.021^{***} (0.005)	-0.202^{***} (0.047)	-0.017^{***} (0.005)
SST	-9.158^{*} (5.409)	-4.972^{*} (2.906)	5.923 (5.161)
SST^2	54.738^{***} (16.038)	-4.608 (7.203)	-56.485^{***} (17.413)
Chlorophyll-a	80.992^{***} (17.061)	-5.366 (11.155)	15.454 (24.835)
Constant	23.745^{***} (2.132)	12.096^{***} (1.639)	28.188^{***} (3.376)
Observations	23	23	23
R^2	0.707	0.608	0.790
Adjusted R^2	0.621	0.493	0.729
Residual Std. Error (df = 17)	7.308	3.917	9.589

Note: *p<0.1; **p<0.05; ***p<0.01.
Robust (Huber–White) SEs via MLR. Biomass in 10^5 tons.
Environmental variables centered at sample means.
Jack mackerel uses Model B interpolation (see Section 3.1).

Allocated harvest enters positively and is highly significant in both fleets, confirming that vessels with larger quota allocations undertake more trips per year. This variable provides the key linkage between the trip equation and the simulation framework: changes in regional TACs translate into vessel-level effort adjustments through the quota share mechanism.

The two fleets differ markedly in their price responsiveness. For the industrial fleet, the jack mackerel price is positive and significant ($p < 0.01$), consistent with this fleet's primary orientation toward jack mackerel. Sardine and anchoveta prices have no detectable effect on industrial effort, reflecting the limited participation of industrial vessels in the sardine-anchoveta fishery. For the artisanal fleet, the sardine price is positive and significant ($p < 0.01$), consistent with sardine being the dominant target species for artisanal vessels in the Centro-Sur region. The anchoveta price enters with a negative and significant coefficient. This counterintuitive sign likely reflects simultaneity: years of low anchoveta availability produce both higher prices (through the inverse demand channel) and fewer trips (through reduced catch opportunities), generating a negative reduced-form correlation. As noted above, the purpose of this equation is not causal identification but rather to provide an empirically grounded mapping for the simulation, where prices are determined endogenously by the inverse demand module.

Hold capacity is not significant for the industrial fleet, where vessels are relatively homogeneous in scale, but is positive and significant for the artisanal fleet. Among artisanal vessels, larger hold capacity is associated with more annual trips, consistent with larger vessels being more commercially active and better able to sustain operations across varying conditions.

The environmental and regulatory variables show distinct patterns across fleets. For the artisanal fleet, adverse weather days are strongly negative and significant ($p < 0.001$), indicating that years with more days of wind speeds above 8~m/s substantially reduce fishing effort for smaller vessels.

For the industrial fleet, adverse weather is not significant, consistent with larger vessels being better equipped to operate under rough conditions. Biological closure days are strongly negative for the industrial fleet, representing the largest effect in the model, consistent with the industrial fleet's dependence on access during open seasons. For the artisanal fleet, the positive coefficient on closure days reflects the current limitation that this variable varies only across regulatory zones (151 days for regions V–VIII versus 182 days for regions IX–XIV) and therefore captures locational differences in fishing activity rather than the causal effect of closures.[3]

Vessel type dummies capture residual heterogeneity in fleet composition. Among artisanal vessels, larger vessel categories (L, BRV) are associated with substantially fewer trips relative to the baseline, reflecting differences in trip duration and operational patterns across vessel classes.

4.3 Climate change projections

The projection methodology is summarized in Section 3.4. Under each scenario and time window, we translate CMIP6-derived environmental deltas into changes in (i) bad weather days entering the trip equation (direct channel) and (ii) species-specific growth capacity entering the SUR equations and, in turn, vessel-level harvest allocations (indirect channel).

4.3.1 Projected environmental changes

Table 3 summarizes the projected environmental changes for the Centro-Sur study area. Sea surface temperature increases by 0.8–2.3°C depending on the scenario, with the strongest warming under SSP5-8.5 end-of-century. Chlorophyll-a declines moderately (2–6%), consistent with increased stratification reducing nutrient supply to the surface. Wind speed changes are negligible across all scenarios (< 0.1 m/s), reflecting the limited sensitivity of near-surface atmospheric circulation to greenhouse forcing in the IPSL model for this region.

Table 3: Projected environmental changes for Centro-Sur Chile (IPSL-CM6A-LR, delta method)

SSP	Window	SST (deg C)	CHL (ratio)	Wind (m/s)
SSP2-4.5	2081–2100	1.480	0.942	0.037
SSP2-4.5	2041–2060	0.811	0.971	0.060
SSP5-8.5	2081–2100	2.333	0.959	0.068
SSP5-8.5	2041–2060	0.970	0.988	-0.049

4.3.2 Projected changes in biomass growth capacity

The SUR environmental coefficients imply asymmetric impacts across species (Table 4). Sardine growth capacity increases substantially under all scenarios, driven by the large positive and signif-

[3]Once year-specific closure dates from annual SUBPESCA resolutions are incorporated, providing within-zone temporal variation, this coefficient is expected to become negative.

icant SST^2 coefficient and the strong CHL response. In contrast, jack mackerel growth declines sharply due to its concave thermal response—warming pushes temperatures away from the species' thermal optimum. Anchoveta growth capacity also declines, though less severely, primarily through the negative SST effect.

Table 4: Projected change in species growth capacity (% change from historical mean, comparative statics on SUR coefficients)

SSP	Window	Sardine	Anchoveta	Jack mackerel
SSP2-4.5	2081–2100	416.6%	-218.3%	-592.5%
SSP2-4.5	2041–2060	108.7%	-88.1%	-167.7%
SSP5-8.5	2081–2100	1111.4%	-463.1%	-1507.6%
SSP5-8.5	2041–2060	169.8%	-115.5%	-243.5%

Note:
Computed at mean biomass levels. Values reflect the partial effect of projected SST and CHL changes on the SUR growth equation, relative to the historical mean of y_i (next-period biomass plus harvest).

4.3.3 Decomposition of climate effects on fishing effort

Table 5 presents the central result: the projected percentage change in annual fishing trips decomposed into the direct weather channel and the indirect biomass channel. The decomposition reveals that the indirect channel dominates—changes in stock productivity, transmitted through harvest allocations, drive the bulk of the effort response.

Table 5: Projected % change in annual fishing trips by channel and fleet

Fleet	SSP	Window	Direct (weather)	Indirect (biomass)	Combined
Artisanal	SSP2-4.5	2081–2100	1.7	153.0	154.6
Artisanal	SSP2-4.5	2041–2060	1.7	20.0	21.7
Artisanal	SSP5-8.5	2081–2100	-4.2	250.7	246.5
Artisanal	SSP5-8.5	2041–2060	-3.0	39.8	36.8
Industrial	SSP2-4.5	2081–2100	0.1	-21.9	-21.8
Industrial	SSP2-4.5	2041–2060	-0.2	-21.6	-21.8
Industrial	SSP5-8.5	2081–2100	-0.1	-21.9	-21.9
Industrial	SSP5-8.5	2041–2060	0.1	-21.7	-21.7

Note:
Direct channel: change in bad weather days from projected wind speed. Indirect channel: change in harvest allocation from projected biomass growth capacity. Combined: both channels applied simultaneously. Harvest allocation scaling capped at [0.2, 3.0] of current levels.

The results reveal a stark asymmetry between fleet segments. Artisanal fishing effort increases by 20–250% across scenarios, driven by the expansion of sardine growth capacity under warming. Since artisanal vessels predominantly target sardine, the increase in sardine biomass translates into larger

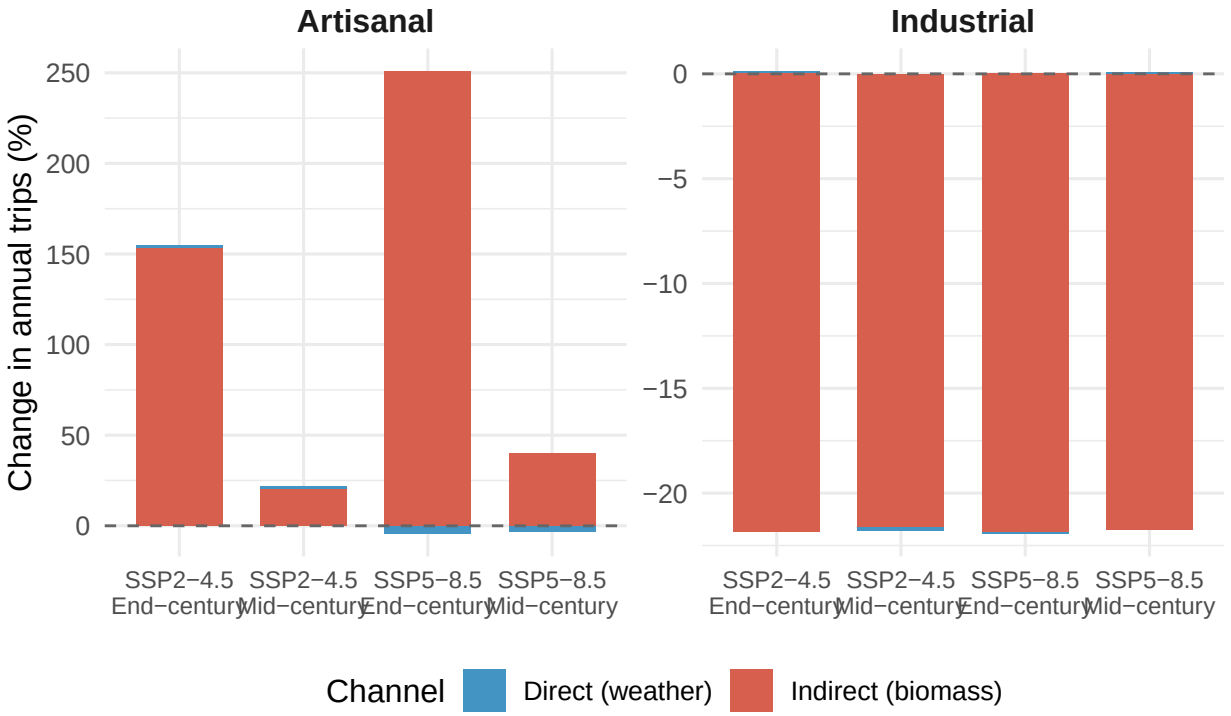

Figure 3: Projected change in annual fishing trips under climate change scenarios, decomposed by channel

harvest allocations and more annual trips. The effect is strongest under SSP5-8.5 end-of-century, where sardine growth capacity more than doubles.

Industrial fishing effort, in contrast, declines by approximately 22% across all scenarios. The industrial fleet is more dependent on jack mackerel, whose thermal optimum response means that warming of any magnitude reduces growth capacity. The consistency of the industrial decline across scenarios reflects the capped harvest allocation scaling and the dominance of jack mackerel in the industrial species mix.

The direct weather channel is negligible in all scenarios (less than 5% in magnitude), reflecting the limited projected changes in wind speed for the Chilean coast under IPSL-CM6A-LR. This finding implies that the primary mechanism through which climate change affects fishing effort in this fishery is biological rather than operational.

These findings have important implications for fisheries management under climate change in Chile. The divergent responses across fleet segments suggest that uniform management policies may be inadequate, and that adaptation planning should account for the heterogeneous vulnerability of different fleets to environmental change.

5 Discussion

Our results reveal a fundamental asymmetry in how climate change affects different segments of Chile's small pelagic fishery. The artisanal fleet, which predominantly targets sardine, stands to benefit from warming through increased stock growth capacity, while the industrial fleet, which depends more heavily on jack mackerel, faces declining productivity as ocean temperatures move away from that species' thermal optimum. This finding aligns with the broader literature on heterogeneous climate impacts across fleet segments (Free et al., 2019; Sumaila et al., 2011) and underscores that aggregate projections can mask important distributional consequences within the same fishery.

The decomposition of climate effects into direct (weather) and indirect (biomass) channels provides a novel contribution to the fisheries-climate literature. The finding that the indirect biomass channel dominates—accounting for virtually all of the projected effort response—has important implications for both research and management. It suggests that the primary pathway through which climate change will reshape fishing activity in this region is through altered species productivity, transmitted via quota allocations, rather than through changes in operational conditions at sea. This result is consistent with the limited sensitivity of near-surface wind patterns to greenhouse forcing in the southeastern Pacific, as projected by the IPSL-CM6A-LR model, although multi-model ensembles would be needed to assess the robustness of this finding.

The near-zero contribution of the direct weather channel ($< 5\%$ of the combined effect) is itself an informative result rather than a null finding to be discounted. It implies that, conditional on the IPSL-CM6A-LR projection of essentially unchanged near-surface winds for the Centro-Sur, climate adaptation policy in this fishery should focus on the institutional architecture that mediates the biomass channel—quota allocation rules, transferability across fleets, and the design of TACs under shifting species productivity—rather than on operational margins such as port infrastructure, vessel reinforcement, or weather-indexed insurance. Investments in operational adaptation are unlikely to mitigate climate exposure as long as the dominant transmission mechanism runs through stock productivity and the rules that translate biomass into vessel-level harvest. This conclusion is conditional on the wind projection used here; if alternative climate models predict larger increases in extreme wind events, the relative weight of the direct channel could rise, particularly for the artisanal fleet.

The species-level biomass projections deserve careful interpretation. The large projected increase in sardine growth capacity under warming reflects the strong positive response of sardine to SST in the estimated SUR equations. While this is consistent with the historical observation that sardine populations in the Humboldt system tend to expand during warm phases (Chavez et al., 2003), the magnitude of the projected changes—particularly under end-of-century SSP5-8.5—should be viewed as indicative of the direction and relative scale of impacts rather than as precise point forecasts. Several sources of uncertainty affect these projections, including the small sample size of

the SUR estimation (N=23), the use of a single climate model, and the assumption that historical relationships between environmental conditions and biomass will remain stable under novel climate states.

The projected decline in jack mackerel growth capacity raises important questions for the management of Chile's largest pelagic fishery. As a highly migratory, transboundary stock managed under the South Pacific Regional Fisheries Management Organisation (SPRFMO), jack mackerel's future trajectory depends not only on local environmental conditions but also on international harvest agreements and the stock's distributional response to warming (Arcos et al., 2001). Our comparative statics approach captures only the local productivity effect and does not account for potential range shifts, which could either exacerbate or mitigate the projected declines depending on whether jack mackerel moves toward or away from Chilean waters.

Several caveats apply to the present analysis. First, the projections hold ex-vessel prices and management rules constant at historical levels, implying that the effort responses capture the mechanical effect of changed environmental and biological conditions without accounting for endogenous price adjustments or adaptive management. Sumaila et al. (2011) note that reduced fish supply under climate change could increase prices, partially offsetting revenue losses from lower catches, while Lam et al. (2016) project sizable changes in global fisheries revenues by the 2050s once price responses and cross-country trade are accounted for. Our companion paper will incorporate an inverse demand system and numerical optimization to capture these equilibrium adjustments. Second, the use of chlorophyll-a projections from a single Earth system model introduces additional uncertainty, as the direction and magnitude of primary productivity changes under climate change remain contested across models (Sumaila et al., 2011). Third, our models do not explicitly incorporate risk preferences or production risk, which may play an important role in shaping fleet-level responses to environmental variability (Kasperski & Holland, 2013; Sethi et al., 2014). Fourth, the biological closure variable in the trip equation currently varies only across regulatory zones (151 days for regions V–VIII and 182 days for regions IX–XIV) and therefore captures locational heterogeneity rather than the causal effect of within-vessel exposure to closed days. This limits our interpretation of the closure coefficient—particularly the counterintuitive positive sign for the artisanal fleet—and is the reason we rely on the quota allocation mechanism, not the closure variable, as the channel through which climate-driven biomass changes affect effort. Incorporating year-specific closure dates from annual SUBPESCA resolutions would introduce within-zone temporal variation and is a planned extension.

The finding that climate change creates winners and losers within the same fishery has direct implications for quota allocation policy. Chile's current system allocates quotas separately for the artisanal and industrial sectors, with limited transferability across fleet segments. If sardine productivity increases while jack mackerel declines, the relative value of artisanal quotas could rise substantially, potentially motivating reforms to the quota allocation system to allow more flexible cross-sector transfers. This echoes the theoretical work of Birkenbach et al. (2020) on multispecies

harvest patterns under catch shares, and the practical observation that secure property rights can help fishers adapt to changing species composition by reallocating effort across their portfolio.

6 Conclusions

This paper estimates a multi-species bioeconomic model for Chile's small pelagic fishery and projects the impact of climate change on fishing effort under CMIP6 scenarios. We contribute to the literature in three ways. First, we jointly estimate stock dynamics and fishing effort in a multi-species setting that captures both biological interdependencies (through a SUR biomass model) and economic responses (through a negative binomial trip equation), applied to a major developing-country fishery. Second, we decompose the projected climate impact on fishing effort into a direct weather channel and an indirect biomass channel, showing that the latter dominates. Third, we document a stark asymmetry between fleet segments: the artisanal fleet benefits from warming-driven sardine expansion while the industrial fleet faces declining jack mackerel productivity, with implications for the distributional consequences of climate change in multi-fleet fisheries.

Our projections indicate that climate change will reshape the composition of fishing activity in Chile's Centro-Sur region. Under moderate warming (SSP2-4.5, mid-century), artisanal effort is projected to increase by approximately 20%, while industrial effort declines by a similar magnitude. Under high-emission end-of-century conditions (SSP5-8.5, 2081–2100), the divergence between fleets becomes substantially larger. These results suggest that management policies focused on aggregate effort levels may miss important heterogeneity across fleets and species, and that climate adaptation in this fishery will need to address the uneven distribution of impacts.

Several extensions to the present framework are planned. A companion paper will incorporate trip-level cost functions and an inverse demand system to enable full numerical optimization of quota paths under climate change, following the complete Kasperski (2015) approach. The spatial dimension of effort allocation is also a natural extension, connecting the multi-species model to the location choice literature (e.g., Dupont, 1993; Hicks et al., 2020; Smith, 2005), since the geographic distribution of species availability is likely to shift under warming. Finally, incorporating risk preferences and production risk—through, for example, random coefficient models—would provide a richer characterization of how fishers respond to the increased environmental variability projected under climate change (Kasperski & Holland, 2013).

7 Repository

The source code for this project is available on GitHub

8 References

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A Additional robustness checks

Table 6: SUR robustness: observed jurel CS only (N = 12)

	Dependent variable:		
	Sardine	Anchoveta	Jack mackerel
	(1)	(2)	(3)
Own biomass (x_{it})	0.733^{***} (0.085)	0.288^{**} (0.120)	0.705^{***} (0.066)
Own biomass^2 (x_{it}^2)	-0.025^{***} (0.002)	-0.289^{***} (0.030)	-0.019^{***} (0.002)
SST	-23.156^{***} (6.761)	-6.703^{*} (3.680)	19.431^{*} (11.587)
Chlorophyll-a	74.023^{***} (15.266)	-30.841^{***} (9.070)	-10.944 (38.159)
Constant	30.089^{***} (1.652)	15.884^{***} (1.359)	35.144^{***} (3.165)
Observations	12	12	12
R^2	0.892	0.540	0.766
Adjusted R^2	0.830	0.277	0.632
Residual Std. Error (df = 7)	6.225	4.697	12.653

Note: *p<0.1; **p<0.05; ***p<0.01. Robust SEs.
Only years with observed (non-interpolated) jurel CS biomass.
SST^2 dropped for parsimony given reduced sample.

Table 7: SUR robustness: two-species, jurel as exogenous (N = 23)

	Dependent variable:	
	Sardine	Anchoveta
	(1)	(2)
Own biomass (x_{it})	0.410^{**} (0.170)	0.666^{***} (0.219)
Own biomass^2 (x_{it}^2)	-0.021^{***} (0.008)	-0.123^{***} (0.037)
SST	-10.159^{*} (5.729)	-1.995 (2.023)
SST^2	53.217^{***} (17.052)	-2.043 (5.831)
Chlorophyll-a	79.958^{***} (16.038)	-2.463 (8.085)
Jack mackerel biomass	-0.064 (0.100)	0.150^{***} (0.040)
Constant	23.911^{***} (2.229)	10.405^{***} (1.259)
Observations	23	23
R^2	0.711	0.745
Adjusted R^2	0.603	0.650
Residual Std. Error (df = 16)	7.485	3.255

Note: *p<0.1; **p<0.05; ***p<0.01. Robust SEs.
Two-species SUR; jack mackerel biomass enters as exogenous covariate.

Table 8: SUR robustness: with cross-species interactions (N = 23)

	Dependent variable:		
	Sardine	Anchoveta	Jack mackerel
	(1)	(2)	(3)
Own biomass (x_{it})	0.426^{***} (0.137)	1.142^{***} (0.163)	1.048^{***} (0.132)
Own biomass^2 (x_{it}^2)	-0.017 (0.010)	-0.202^{***} (0.054)	-0.021^{***} (0.005)
SST	-9.969^{*} (5.850)	-4.172 (2.935)	6.055 (4.696)
SST^2	52.621^{***} (16.586)	-1.582 (7.976)	-54.858^{***} (16.153)
Chlorophyll-a	80.145^{***} (17.796)	-2.248 (14.017)	5.593 (24.541)
Sardine × Anchoveta	-0.0001 (0.022)	-0.019 (0.017)	
Sardine × Jack mackerel	0.007 (0.011)		-0.014^{*} (0.008)
Anchoveta × Jack mackerel		-0.010 (0.023)	0.041^{**} (0.020)
Constant	24.183^{***} (2.420)	12.108^{***} (1.551)	26.270^{***} (3.477)
Observations	23	23	23
R^2	0.711	0.618	0.814
Adjusted R^2	0.575	0.440	0.728
Residual Std. Error (df = 15)	7.738	4.116	9.601

Note: *p<0.1; **p<0.05; ***p<0.01. Robust SEs.
Main specification augmented with pairwise biomass interactions.
No interaction term is individually or jointly significant.